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Navigating the Complexities of Qualitative Comparative Analysis: Case Numbers, Necessity Relations, and Model Ambiguities

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Abstract

Background: In recent years, the method of Qualitative Comparative Analysis (QCA) has been enjoying increasing levels of popularity in evaluation and directly neighboring fields. Its holistic approach to causal data analysis resonates with researchers whose theories posit complex conjunctions of conditions and events. However, due to QCA's relative immaturity, some of its technicalities and objectives have not yet been well understood. **Objectives:** In this article, I seek to raise awareness of six pitfalls of employing QCA with regard to the following three central aspects: case numbers, necessity relations, and model ambiguities. Most importantly, I argue that case numbers are irrelevant to the methodological choice of QCA or any of its variants, that necessity is not as simple a concept as it has been suggested by many methodologists, and that doubt must be cast on the determinacy of virtually all results presented in past QCA research.

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Method: By means of empirical examples from published articles, I explain the background of these pitfalls and introduce appropriate procedures, partly with reference to current software, that help avoid them. **Conclusion:** QCA carries great potential for scholars in evaluation and directly neighboring areas interested in the analysis of complex dependencies in configurational data. If users beware of the pitfalls introduced in this article, and if they avoid mechanistic adherence to doubtful "standards of good practice" at this stage of development, then research with QCA will gain in quality, as a result of which a more solid foundation for cumulative knowledge generation and well-informed policy decisions will also be created.

Keywords
configurational comparative methods, csQCA, fsQCA, mvQCA, QCA, Qualitative Comparative Analysis

Introduction

Evaluation has been one of the fastest-growing areas of the last years in which Qualitative Comparative Analysis (QCA)—an empirical social research method largely popularized by Charles Ragin (1987, 2000, 2008)—has been making inroads. To date, at least 19 publications can be identified in peer-reviewed English-language journals. In the group of "agenda articles," Befani (2013), Befani, Ledermann, and Sager (2007), Blackman, Wistow, and Byrne (2013), Sager and Andereggen (2012), Verweij and Gerrits (2013), and Warren, Wistow, and Bambra (2014) showcase the usefulness of QCA for addressing various research questions evaluators face. In contrast, "applied articles" focus on empirical substance and include Balthasar (2006), Blackman (2008), Chatterley et al. (2014), Dy et al. (2005), Ford, Duncan, and Ginter (2005), Glatman-Freedman et al. (2010), Gross and Garvin (2011), Holvoet and Dewachter (2013), Kahwati et al. (2011), Lam and Ostrom (2010), Ledermann (2012), Ozegowski (2013), and Thygeson et al. (2012). A wide range of topics has been covered by these works, from the determinants of the evaluation type that state agencies use, over the effectiveness of public–private partnership contracts for toll roads, to the quality of medical homes. Table 1 provides an overview, including the area from which applied studies come, the topic they have dealt with and the QCA variant they have employed.[1]

Table 1. Studies Published in the Area of Evaluation.

Study	Category	Area	Topic	Variant
Balthasar (2006)[a]	Applied	Administration	Evaluation use	mvQCA
Befani (2013)	Agenda	Methodological	—	—
Befani et al. (2007)	Agenda	Methodological	—	—
Blackman (2008)	Applied	Public health	Smoking cessation	csQCA
Blackman, Wistow, and Byrne (2013)	Agenda	Public health	—	—
Chatterley et al. (2014)[a]	Applied	Public health	School sanitation	fsQCA
Dy et al. (2005)	Applied	Public health	Pathway effectiveness	csQCA
Ford et al. (2005)	Applied	Public health	Health agencies	csQCA
Glatman-Freedman et al. (2010)	Applied	Public health	Vaccine introduction	csQCA
Gross and Garvin (2011)[a]	Applied	Infrastructure	Toll roads	mvQCA
Holvoet and Dewachter (2013)[a]	Applied	Development	Evaluation societies	csQCA
Kahwati et al. (2011)[a]	Applied	Public health	Obesity	csQCA
Lam and Ostrom (2010)[a]	Applied	Development	Irrigation systems	csQCA
Ledermann (2012)[a]	Applied	Administration	Evaluation use	csQCA
Ozegowski (2013)[a]	Applied	Public health	Primary care	fsQCA
Sager and Andereggen (2012)	Agenda	Methodological	—	—
Thygeson et al. (2012)	Applied	Public health	Medical homes	fsQCA
Verweij and Gerrits (2013)	Agenda	Methodological	—	—
Warren et al. (2014)	Agenda	Methodological	—	—

Source. www.compasss.org/bibdata.htm, accessed July 3, 2014.
Note. QCA = Qualitative Comparative Analysis; mvQCA = multi-value QCA; csQCA = crisp-set QCA; fsQCA = fuzzy-set QCA.
[a] Data available.

Against the backdrop of rising publication figures, the objective of this article is to raise awareness of common pitfalls of QCA and to offer advice on how to avoid them. Among the vast array of aspects that create stumbling blocks, three major ones are singled out because of their ubiquity. First, the *number of cases* has often been cited to rationalize the choice of QCA in preference to other techniques of empirical data analysis. I prove this justification to rest on misconceptions about the comparative advantages of different methods and show why case numbers are likewise irrelevant for determining the variant of QCA. Second, I argue for a radical rethink of current policy concerning the treatment of *necessity relations*. Unlike the search for minimal sufficiency, strategies for identifying minimal necessity relations that are not simultaneously part of sufficiency analyses have so far been limited to trial-and-error assessments of simple conditions. Furthermore, I argue that the concept of necessity has been dissociated from QCA's "analytical moment" unduly. Third, I demonstrate that authors of QCA studies have not reported *model ambiguities* or have not even been cognizant of their presence in the first place. The sources of these pitfalls reside in the undocumented selection of models on the mere basis of theoretical preferences and software deficiencies stemming from an unquestioned reliance on Quine–McCluskey (QMC) optimization for purposes of causal data analysis.

Data material from many applied articles is used in order to illustrate these points, whose order is not random. Rather, it reflects the severity of the pitfalls associated with each of the three aspects. The citation of case numbers is relatively inconsequential for the actual application of QCA itself and the problematic treatment of necessity relations has some negative downstream effects but limitedly so. In contrast, the unknown existence of model ambiguities due to the misapplication of QMC and the widespread practice of preference-based model selection cannot be described other than as disastrous from the perspective of cumulative knowledge generation.

A caveat on the article's target audience: As QCA has reached a critical mass of publications with more than 100 journal articles per year in 2013 and more than 500 articles in total since Ragin, Mayer, and Drass (1984), I will not explain the principal ideas of this method again. At a minimum, readers should be familiar with the essentials of crisp-set QCA (csQCA), multi-value QCA (mvQCA), and fuzzy-set QCA (fsQCA), as introduced in the edited volume by Rihoux and Ragin (2009).[2] The target audience of this article is thus readers with at least an intermediate knowledge of QCA. However, those with a basic knowledge will still be able to follow the subsections on Pitfalls 1a and 1b, 2a, and 3a. For reasons of comparability, I do not use a consistent syntax but adhere to the conventions of the original publications which I refer to.

Navigating Pitfalls in Research With QCA

In the following subsections, I elaborate on six pitfalls of QCA, which relate to the following three different aspects: *case numbers*, *necessity relations,* and *model ambiguities*.

Aspect 1: Case Numbers

The number of cases has often been cited as a criterion on which the choice of QCA in general and that of any of its variants in particular should be based. More precisely, it has been suggested that small numbers of cases increase the advantage of QCA over "quantitative" methods such as regression analysis, and medium numbers over "qualitative" methods such as case studies. Within the family of QCA, similar distinctions have been drawn with regard to csQCA, mvQCA, and fsQCA.

Pitfall 1a: Choosing QCA on the basis of case numbers. The position of QCA vis-à-vis other research methods has often been specified in relation to case numbers. For example, Wagemann and Schneider (2010, p. 377) hold that its full potential "unfolds in studies based on a mid-sized *N*. This is precisely why QCA is an appealing alternative for those scholars whose number of cases [. . .] do not permit either a standard statistical analysis or a classical comparative case study approach." Rihoux (2003, p. 353) agrees that "QCA is particularly well suited for small-*n* studies" with case numbers from "fourteen to forty-four," as do Ragin, Shulman, Weinberg, and Gran (2003, p. 324) who emphasize that QCA "offers an approach well suited for the examination of a moderate number of cases [. . .] too few for the application of commonly used multivariate statistical techniques [. . .] but too many for in-depth, case-oriented analysis."

Evaluation researchers have often misread these statements as confirmation that "QCA is only capable of comparing a limited number of cases" (Befani, Ledermann, & Sager, 2007, p. 175; see also Kahwati et al., 2011, p. 458; and Sager & Andereggen, 2012, p. 65). However, should the broader conclusion also be that QCA represents the obvious choice if researchers have more cases than could be handled within the confines of case studies but not so many that regression techniques suggest themselves? The answer is an unequivocal "no." Theoretically, QCA proper works with any number of cases larger than one as long as they differ at least minimally on conditions and outcome. A parallel minimum requirement—at least two cases and variation on covariates—exists for regression analysis. Similarly, case studies with two cases or more are not unusual.[3] The number of cases is therefore an ill-suited criterion for judging the appropriateness of QCA vis-à-vis other methods of social–scientific inquiry. This of course does not challenge the view of many methodologists that an intimate knowledge of cases is a *sine qua non* for the interpretation of results (Rihoux, 2003, p. 360), which remain holistic and case based, but case knowledge *per se* is not strictly required at any stage of the analysis. To conclude, the choice of QCA should be justified on the basis of researchers' interest in configurational analysis, and not on the basis of case numbers.

Pitfall 1b: Choosing variants of QCA on the basis of case numbers. Provided QCA has been identified as the preferred method, Rihoux (2006, p. 686f) and Herrmann and Cronqvist (2009) claim that csQCA is geared toward small numbers of cases, fsQCA toward large numbers, and mvQCA toward medium numbers. Researchers in evaluation have largely adopted these views uncritically, arguing that "since the advent of fsQCA, it has become obvious that, in addition to small *n*, QCA can handle a medium as well as a large number of cases" (Befani, 2013, p. 278).[4]

Indubitably, QCA is a case-based method but case orientation does not imply that csQCA is unsuitable for large-*n* contexts, just as mvQCA is not the default option for medium-*n* designs. Similar to the comparative advantages of different research methods, the conditioning of variants on case numbers lacks a formal rationale. No first principles exist on the basis of which the correct QCA variant could be deduced from the number of cases in hand. Instead, it is mechanically determined by the nature of the sets formed by the values of condition and outcome factors. If among the frame of condition factors, all of which are crisp, at least one factor is multivalent then there is no way around mvQCA, irrespective of the number of cases. If among the frame of factors, all of which are bivalent, at least one is fuzzy, then the use of fsQCA is unavoidable. Finally, if all factors are bivalent and their values should generate crisp sets then csQCA has to be employed.[5] Case numbers are extraneous to the decision on the types of sets to be used and, by extension, the variant of QCA.

Aspect 2: Necessity Relations

According to "standards of good practice in QCA" assembled by some methodologists, "it is always recommended that analyses of necessity and sufficiency be kept separate," "with the analysis of necessary conditions going first" because "any inference about the presence or absence of necessary conditions based on the top-down logical minimization of truth tables is prone to produce flawed results" (Schneider & Wagemann, 2010, p. 8; 2012, pp. 105, 114).[6] In these and similar warnings, necessary conditions are generally understood to be simple conditions or disjunctions of simple conditions forming substitutable parts of higher order concepts that are deemed theoretically plausible (Schneider & Wagemann, 2012, p. 80).

Unfortunately, the authoritative resoluteness exhibited by these scholars in declaring what counts as "good practice" with regard to the analysis of necessity and what not masks important facts that disappear in generalizations of worrying inaccuracy. In relation to the concept of necessity, at least two objections must be raised: First, the analysis of minimally necessary conditions for an outcome without concurrent regard to its sufficient conditions should not be confined to simple conditions or be driven by theoretical expectations but by a neutral and systematic procedure capable of bringing all data-fitting expressions to the attention of the researcher.[7] Second, the dissociation of necessity analyses from what Schneider and Wagemann (2012, p. 11) refer to as QCA's "analytic moment"—the phase from the construction of the truth table over the process of logical minimization to the identification of the solution—distorts the central meaning of the concept of necessity in QCA.

Pitfall 2a: Restricting isolated analyses of minimal necessity relations to simple conditions. Following the instructions provided in "standards of good practice in QCA," typical analyses of isolated necessity relations have so far been limited to user-driven explorations with simple conditions or theoretically preconceived disjunctions of simple conditions by means of *fs/QCA* (Ragin & Davey, 2014)—the most popular computer program with a market share upward of 80% (Thiem & Duşa, 2012, p. 45).

With the goal of enhancing the search for such minimal necessity relations, Bol and Luppi (2013) have proposed an algorithm, their so-called *systematic necessity assessment*, which consists of three steps. First, for each observed truth table configuration, AND is replaced by OR and the necessity of the resulting disjunctions is tested with respect to the outcome. If this pretest returns no consistently necessary conditions, the search can be aborted because no disjunction of lower complexity will turn out necessary.[8] Second, necessity consistency for each simple condition is computed if any one disjunctive configuration has proven to be necessary. For simple conditions that have not been identified as necessary, all remaining conditions are added one by one to each inconsistently necessary condition until consistent relations of necessity materialize.

Although this approach offers a significant improvement, it still stops short of exhaustive systematization and has the disadvantage of requiring a truth table. A more efficient strategy is proposed by Thiem and Duşa (2013c) in their *super-/subset analysis*, which is implemented in the authors' *QCA* package for the *R* environment (Duşa & Thiem, 2014; *R* Development Core Team, 2014; Thiem & Duşa, 2013b). The starting point is the set of all simple conditions, whose size is determined by the number of values of each condition factor. For example, if there are two bivalent factors and one trivalent factor, the size of the initial set to be tested amounts to $2 + 2 + 3 = 7$ conditions. If any of these simple conditions already proves necessary, further simple conditions are added to form more complex conjunctions until necessity is lost. If a simple condition turns out not to meet the preset minimum of consistency, further simple conditions are added disjunctively until this level is obtained. The algorithm thus takes a different starting point and considers both disjunctions and conjunctions in contrast to Bol and Luppi's procedure, whereby it is ensured that all expressions which meet the criteria set by the researcher will be revealed. Besides the *QCA* package, only the standalone program *Kirq* offers similar functionality (Reichert & Rubinson, 2014; Rubinson, 2013), but unlike the latter, *QCA* is capable of handling multivalent condition factors, for which it also provides consistency and coverage statistics (Thiem, 2014a).

I demonstrate the advantages of the super-/subset algorithm by rerunning isolated analyses of necessity for all applied articles listed in Table 1, which have performed explicit tests of necessity and for which data were available. The results are presented in Table 2. The first column gives the reference, "Out." the outcome in question, the third column the consistency cut-off that was originally used by the authors, followed by the number of conditions that were found to meet this cut-off, the number of conditions identified by *QCA* at this cut-off, and the number of conditions identified at a new consistency cut-off of 0.75, which is usually recommended by methodologists as the minimum for sufficiency tests (Ragin, 2008, p. 46; Schneider & Wagemann, 2012, p. 279).[9]

While only Chatterley et al. (2014) have found a single condition for the outcome of well-managed school sanitation services, which was a disjunction of two simple conditions, but none for the negation of this outcome, the reanalysis reveals that there exist no fewer than 30 conditions that are minimally necessary for the outcome and 35 for its negation. At a consistency cut-off of 0.75, 27, and 30 conditions, respectively, are found. In the case of Ozegowski (2013), who finds no condition to be necessary for the outcome of equity in the geographical distribution of general practitioners in Europe, the reanalysis identifies 51 conditions when the consistency cut-off is set to the original value of 0.9, and 41 conditions when the cut-off is lowered to 0.75.

Table 2. Reanalysis of Necessity Tests for Four Evaluation Studies.

Study	Outcome	Cons. Cut-off (Orig.)	Number of Conditions (Orig.)	Number of Conditions	Number of Conditions (Cons. 0.75)
Chatterley et al. (2014)	$SSS^{\{1\}}$	0.90	1	30	27
	$SSS^{\{0\}}$	0.90	None	35	30
Holvoet and Dewachter (2013)	$NE^{\{1\}}$	0.90	None	20	21
	$NE^{\{0\}}$	0.90	None	20	24
	$AC^{\{1\}}$	0.90	None	20	16
	$AC^{\{0\}}$	0.90	None	18	13
Ledermann (2012)	$CD^{\{1\}}$	? (1.0)	None	6	8
	$CD^{\{0\}}$	? (1.0)	None	6	8
Ozegowski (2013)	$EQ^{\{1\}}$	0.90	None	51	40
	$EQ^{\{0\}}$	0.90	None	31	36

Note. Cons. = consistency; Orig. = original study; $SSS^{\{1/0\}}$ = well-managed/not well-managed school sanitation services; $NE^{\{1/0\}}$ = networkability/no networkability; $AC^{\{1/0\}}$ = accountability/no accountability; $CD^{\{1/0\}}$ = change decision/no change decision; $EQ^{\{1/0\}}$ = equity/no equity.

These results show clearly that if researchers only explore simple conditions or *a priori* plausible disjunctions as recommended by Schneider and Wagemann (2012), the vast majority of data-fitting conditions will be missed, but an *a posteriori* search for higher-order concepts that happen to be theoretically plausible to the analyst seems to be of equally dubious scientific benefit. The logical conclusion can only be the following: when too many minimally necessary conditions meet all formal criteria, any attempts at substantive interpretation should be aborted. More generally, however, the role of such isolated necessity tests for purposes of causal data analysis within the context of QCA remains unclear.[10]

Pitfall 2b: Dissociating necessity relations from QCA's "analytical moment". Methodologists have also argued that relations of necessity have no place in QCA's analytical moment. For instance, Schneider and Wagemann (2012, p. 105) claim that "[a] truth table [. . .] does not play an important role in the analysis of necessity" and that "statements about necessity should only be made if necessary conditions have been identified in a separate and adequate analysis" (Schneider & Wagemann, 2010, p. 8). These authors also argue that necessary conditions which involve disjunctions should only be interpreted if the separate disjuncts form "functional equivalents," that is, substitutable parts of some higher-order concept (Schneider & Wagemann, 2012, p. 74).[11] Unfortunately, such statements have put researchers under the misapprehension that relations of necessity in QCA are not only generally limited to simple conditions but also conceptually unconnected to the method's analytical moment. The first misconception has been partly addressed earlier; the second will be corrected in this section.

It is a widely recommended practice in QCA to add or replace condition factors in order to resolve inconsistencies in the truth table before minimization is performed (Marx, Cambré, & Rihoux, 2013, p. 29; Rihoux & De Meur, 2009, p. 48; Schneider & Wagemann, 2010, p. 9). Any attempt at resolving such inconsistencies, however, has the effect of increasing sufficiency coverage with respect to the outcome under investigation and thus amounts to a search for complex necessary conditions. If no inconsistencies exist in the first place, a necessary condition consisting of sufficient yet still non-minimal conditions for the outcome has already been found. During the process of Boolean minimization, this condition is transformed into a complex necessary condition consisting of minimally sufficient conditions before it is separated into sets of minimally sufficient conditions that together also form a minimally necessary condition for the outcome. Contrary to what has been suggested by most representatives of QCA, necessity relations thus do form part of the method's analytical moment, and very centrally so.

To illustrate how the concept of necessity pervades the analysis of sufficiency and because these authors have followed the above-mentioned recommendations before constructing their truth tables, I draw on the work by Gross and Garvin (2011) on the effectiveness of public–private partnership contracts for toll roads. Although a total of three outcome factors are introduced, I focus only on the achievement of a specific toll rate (TORT; $0 = $ no, $1 = $ yes). The set of five condition factors in this mvQCA study comprises the toll rate approach (PRIC; $0 =$ average cost pricing, $1 =$ marginal social-cost pricing, and $2 =$ revenue-maximizing pricing), the concession length ($0 =$ variable, $1 =$ short, and $2 =$ long), upside revenue sharing ($0 =$ absent and $1 =$ present), downside risk sharing ($0 =$ absent and $1 =$ present), and the traffic-demand risk (RISK; $0 =$ low and $1 =$ high). The truth table with respect to the achievement of a specific toll rate, TORT$\{1\}$, is presented in Table 3.

Using the configuration index c_i, it is obvious that the disjunction of conjunctions c_1 to c_8 is perfectly sufficient for the outcome because within none is a case associated with its negation. Expression E_1 in Equation 1 formulates this relation.

$$E_1 : c_1 + c_2 + c_3 + c_4 + c_5 + c_6 + c_7 + c_8 \rightarrow \mathrm{TORT}\{1\} \tag{1}$$

Table 3. Truth Table for Data From Gross and Garvin (2011).

c_i	PRIC	LENG	UPSI	DOSI	RISK	Out(TORT{1})	Consistency	n
1	0	0	0	0	0	1	1	1
2	0	0	0	1	1	1	1	1
3	0	1	0	0	0	1	1	1
4	0	1	0	0	1	1	1	1
5	0	1	0	1	1	1	1	1
6	1	2	1	1	0	1	1	1
7	2	0	0	1	0	1	1	1
8	2	2	0	0	1	1	1	1
9	1	1	0	0	0	0	0	1
10	1	1	0	0	1	0	0	1
11	1	1	1	0	0	0	0	1
12	1	2	1	0	0	0	0	1
13	2	0	1	0	1	0	0	1
14	2	1	0	0	1	0	0	2
15	2	1	1	0	1	0	0	1
16	2	2	0	0	0	0	0	2

Moreover, there is no case in Table 3 such that the outcome is associated with the negation of the disjunction of configurations, which implies that the former is also perfectly sufficient for the latter, as formulated by expression E_2 in Equation 2. In other words, the disjunction of configurations is necessary for the outcome.

$$E_2 : \mathrm{TORT}\{1\} \rightarrow c_1 + c_2 + c_3 + c_4 + c_5 + c_6 + c_7 + c_8 \tag{2}$$

In Boolean algebra, a biconditional dependency integrating Equations 1 and 2, for instance, is called an equivalence and typically uses the bidirectional operator "$\leftrightarrow$."[12] As a result, expressions E_1 and E_2 can be integrated into expression E_3 in Equation 3.

$$E_3 : c_1 + c_2 + c_3 + c_4 + c_5 + c_6 + c_7 + c_8 \leftrightarrow \mathrm{TORT}\{1\} \tag{3}$$

Once all prime implicants (PIs) have been derived by means of logical minimization, the result is expression E_4 given in Equation 4, which represents a non-minimally necessary condition of minimally sufficient conditions for the outcome.[13]

$$\begin{aligned} E_4 : &\ \mathrm{DOSI}\{1\} + \mathrm{PRIC}\{0\} + \mathrm{LENG}\{0\}\mathrm{RISK}\{0\} + \mathrm{LENG}\{0\}\mathrm{UPSI}\{0\} \\ &+ \mathrm{LENG}\{2\}\mathrm{RISK}\{1\} \leftrightarrow \mathrm{TORT}\{1\} \end{aligned} \tag{4}$$

Table 4. Parsimonious Solution for Outcome INST(1).

Prime Implicant	Consist.	R. Cover.	U. Cover.	M^*	M_1
PURP(0)	1.000	0.600	0.600	0.600	0.600
DIST(1)PURP(1)	1.000	0.200	0.200	0.200	0.200
DIST(0)PURP(2)	1.000	0.200	0.000	0.200	
PURP(2)ROUT(0)USEF(1)	1.000	0.200	0.000		0.200
M^*	1.000	1.000			
M_1	1.000	1.000			

Note. Consist. = consistency; R. Cover. = raw coverage; U. Cover. = unique coverage.

In order to transform E_4 into a minimally necessary condition, all redundant PIs have to be eliminated. As can be seen in the truth table in Table 4, DOSI$\{1\}$ covers c_2, c_5, c_6, and c_7, PRIC$\{0\}$ c_3 and LENG$\{2\}$RISK$\{1\}$ covers c_4 and c_8. Once these three have been chosen, all other PIs can be dropped from E_4. The end product is E_5 given in Equation 5—a minimally necessary condition of the outcome consisting of minimally sufficient conditions.[14]

$$E_5 : \text{DOSI}\{1\} + \text{PRIC}\{0\} + \text{LENG}\{2\}\text{RISK}\{1\} \leftrightarrow \text{TORT}\{1\} \tag{5}$$

In summary, contrary to what "standards of good practice in QCA" suggest, necessity relations form an essential part of QCA's analytical moment. For example, any strategy that aims at reducing inconsistencies amounts to a search for complex necessary conditions for the outcome.

Aspect 3: Model Ambiguities

Partly due to tacit model selection, partly to uncritical reliance on QMC, the results presented in a considerable number of publications suggest far clearer patterns in the data than there really are. At best, biased summaries of previous findings are the only repercussions; at worst, costly yet ineffective policies will be implemented following research-based recommendations.[15] The subject of model ambiguities has already been broached elsewhere in relation to a sociological study of Norwegian grassroots organizations (Thiem & Duşa, 2013a), but the problem is not confined to specific areas. Studies in evaluation are as likely to be affected as those in sociology or any other discipline. Although the ignorance of model ambiguities represents one of the severest pitfalls in research with QCA, neither the problem nor its full extent have been recognized so far.

Pitfall 3a: Selecting models tacitly on the basis of theoretical preferences. When *fs/QCA* outputs multiple alternative PIs, researchers are expressly advised to perform a selection in accordance with theoretical knowledge (Ragin, Strand, & Rubinson, 2008, p. 65). As users typically interpret this advice as official sanction for *ad hoc* model choice, neither the extent of ambiguity that surrounds the presented model nor the selection criteria are usually documented. This widespread practice, however, ignores minimal standards of transparency. As Rihoux, Ragin, Yamasaki, and Bol (2009, p. 168) suggest, "[i]f there are many possible minimal formulas [. . .] the choice of a specific minimal formula should be well-documented and justified." Researchers have usually disregarded these conventions, perhaps partly because they receive no mention in more recent textbooks (e.g., Schneider & Wagemann, 2012). The following example is indicative.[16]

Holvoet and Dewachter (2013) seek to identify the factors that influence the performance of national evaluation societies in 37 low- and middle-income countries. Under the heading of "context" they include increased political openness (O) and increased government interest (G), while the set of actor-related factors comprises the availability of financial resources (E), a high degree of activity (A) and extensive membership (M). Using the programs *Tosmana* (Cronqvist, 2011) and *fs/QCA* (Ragin & Davey, 2014), the authors claim to have identified the complex solution for the outcome of high performance on stimulating networking as model M_1 given in Equation 6.

$$\begin{aligned} M_1 : \mathrm{OGeM} + \mathrm{OGea} + \mathrm{OgEAM} + \mathrm{oeA} + \mathrm{oAm} \\ + \mathrm{oEaM} \rightarrow \mathrm{NETWORKING} \end{aligned} \tag{6}$$

What the authors do not report is a second model which both *Tosmana* and *fs/QCA* identify in addition to Equation 6.[17] This model, labeled M_2, is given in Equation 7 and includes the PI AGeM instead of OGeM.

$$\begin{aligned} M_2 : \mathrm{AGeM} + \mathrm{OGea} + \mathrm{OgEAM} + \mathrm{oeA} + \mathrm{oAm} \\ + \mathrm{oEaM} \rightarrow \mathrm{NETWORKING} \end{aligned} \tag{7}$$

The two paths only differ by a single term, but each conjunct emphasizes the relevance of a very different factor. Condition O stresses the role of context, whereas condition A that of actor attributes. In this deliberately simple example, models M_1 in Equation 6 and M_2 in Equation 7 are maximally similar, which would have implied minimal consequences for the interpretation of the overall solution. However, this is not always the case as I show in the following subsection. In conclusion, solution transparency should always be ensured by either listing all data-fitting models or at least reporting the exact degree of ambiguity that surrounds a chosen model, for whose selection, if warranted, well-founded criteria should be cited.

Pitfall 3b: Employing QMC optimization for causal data analysis. QMC has been considered to be QCA's central feature, but it has usually been associated only with the initial phase in which the laws of distribution and complementarity are invoked to derive the PIs (e.g., Longest & Vaisey, 2008, p. 82; Ragin, 2008, p. 135; Schneider & Wagemann, 2012, pp. 104–115). In fixating on this phase, it has been overlooked that some of the algorithm's later mechanisms create serious pitfalls for the analysis of causal structures, which is the objective of virtually all applications of QCA in the social sciences (cf. Schneider & Wagemann, 2012, pp. 8, 313). To understand how the solution outputs by QMC-based software such as *fs/QCA* (Ragin & Davey, 2014), *fuzzy* (Longest & Vaisey, 2008), and *Tosmana* (Cronqvist, 2011) are produced, however, it is unavoidable to transcend the bare process of identifying PIs.

Developed at the intersection of propositional logic and switching circuit theory, QMC's search target is a minimal sum equivalent of the function that describes the truth table, where "sum" simply refers to the Boolean OR operation (McCluskey, 1956; Quine, 1952).[18] Its focus on minimal sums thus implies that a premium is placed on models with minimal equifinality. Being purpose-built to this end, QMC is undoubtedly appropriate for simplifying propositions so as to render them more transparent or for designing electrical switching circuits at minimum cost yet ill-suited to causally oriented analyses of configurational data. The simple reason for this unsuitability is that causal structures need not follow the logic of minimal equifinality. In order to substantiate this claim, I draw on Balthasar's (2006) study about the impact of institutional design on the use types of evaluations—the first application of mvQCA with the *Tosmana* software.[19]

The author analyzes 10 cases of evaluations undertaken in the Swiss Federal Administration between 1999 and 2002 to identify the impact of four condition factors on five outcomes describing the type of an evaluation's final use by the evaluee. The four condition factors are the institutional distance between the evaluator and the evaluee (DIST; 0 = small and 1 = large), the basic evaluation objectives (PURP; 0 = formative, 1 = summative, and 2 = formative and summative), whether the evaluation was carried out in a context where evaluation is routine (ROUT; 0 = no and 1 = yes) and whether the evaluation was geared toward the information requirements of the evaluee (USEF; 0 = no and 1 = yes). I limit the replication to the outcome defining the instrumental use of evaluations by the evaluee (INST, 0 = no and 1 = yes), whereby "impetus that is intentionally incorporated into the policy formulation processes in a manner that can be proven" is referred to (p. 354).[20]

Assume for the sake of argument that we have complete information about the process that generated the data collected by Balthasar, who has no such information. More specifically, assume we know that the institutional use of an evaluation results if, and only if, (1) the evaluee pursues formative goals or (2) a large institutional distance between the evaluator and the evaluee exists and the evaluee pursues summative goals or (3) formative and summative goals are pursued, evaluation is routine and geared toward the information requirements of the evaluee. Model M* given in Equation 8 would describe this data-generating process.

$$\begin{aligned} M^{*} : \; &\mathrm{PURP}(0) + \mathrm{DIST}(1)\mathrm{PURP}(1) + \mathrm{PURP}(2)\mathrm{ROUT}(1)\mathrm{USEF}(1) \\ &\leftrightarrow \mathrm{INST}(1) \end{aligned} \tag{8}$$

The solution Balthasar (p. 365) presents, however, consists of model M_1 given in Equation 9.

$$\begin{aligned} M_1 : \; &\mathrm{PURP}(0) + \mathrm{DIST}(1)\mathrm{PURP}(1) + \mathrm{DIST}(0)\mathrm{PURP}(2) \\ &\leftrightarrow \mathrm{INST}(1) \end{aligned} \tag{9}$$

The pursuit of formative goals by the evaluation and the conjunction of a large distance between evaluator and evaluee and the pursuit of summative goals are common to M* in Equation 8 and M_1 in Equation 9, but the third path differs. In Equation 9, it is the conjunction of a short institutional distance between evaluator and evaluee and the pursuit of formative as well as summative goals. In contrast to above-mentioned Pitfall 3a, where Holvoet and Dewachter (2013) knew about the existence of a second model but failed to report it, *Tosmana* did not even output M*. Are the summary statistics of M* worse than those of M_1?

All model parameters of fit are presented in Table 4, where the two *essential* PIs—those paths to the outcome shared across M* and M_1—are listed in the upper part, and the *inessential* PIs—those paths specific to each model—in the lower part. Consistency and coverage figures prove that no model fits the data worse than the other. When both inessential PIs are put into the same model, their unique coverage becomes zero, which means that redundancies exist, but only one can be dropped if the model is to remain entailed by the truth table. In this case, the unique coverage of the remaining inessential PI increases from 0.0 to 0.2. To summarize at this point, there are no formal grounds for preferring one model to the other. Each reflects the configurational patterns contained in the data. Although the data-generating model M^* in Equation 8 was not output by the software, at least two of the three paths, which together already account for 80% of the cases, have been correctly identified.

Now consider the analysis of the absence of an institutional use as the outcome. Assume again that we can be certain about the data-generating process. An institutional use is absent if, and only if, (1) the institutional distance is large and both formative and summative goals are pursued, (2) the evaluee only pursues summative goals and evaluation is not routine, or (3) the institutional distance is small, evaluation is routine and geared toward the requirements of the evaluee. This model, denoted by $M^{\#}$, is presented in Equation 10.

$$\begin{aligned} M^{\#} : &\ \mathrm{DIST}(1)\mathrm{PURP}(2) + \mathrm{PURP}(1)\mathrm{ROUT}(0) \\ &+ \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \end{aligned} \tag{10}$$

Although Balthasar did not present this analysis, he would have found model M_1 given in Equation 11 had he performed it with the help of *Tosmana*, and only this model.

$$M_1 : \mathrm{DIST}(1)\mathrm{PURP}(2) + \mathrm{DIST}(0)\mathrm{PURP}(1) \leftrightarrow \mathrm{INST}(0) \tag{11}$$

This time, the sole conjunction common to both models is that of a large distance between evaluator and evaluee and the pursuit of both formative and summative goals. As before, all summary statistics are the same.[21] To recapitulate at this point, no formal grounds for preferring one model over the other exist. Each reflects the patterns in the data, but model $M^{\#}$ was not found by the software. At least, however, one path has been correctly identified.

We are still in the enviable position to have complete information about the data-generating model, but what happens if this untenable assumption is dropped? While only two equally well-fitting models with respect to the presence of an institutional use of evaluation actually exist, namely M^* in Equation 8 and M_1 in Equation 9, the solution with respect to the negation of this outcome consists of no fewer than 10 models! These are listed as M_1 to M_{10} in Equations 12 to 21.

$$M_1 : \mathrm{DIST}(0)\mathrm{PURP}(1) + \mathrm{DIST}(1)\mathrm{PURP}(2) \leftrightarrow \mathrm{INST}(0) \tag{12}$$

$$\begin{aligned} M_2 : &\ \mathrm{DIST}(0)\mathrm{PURP}(1) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{ROUT}(0)\mathrm{USEF}(0) \\ &\leftrightarrow \mathrm{INST}(0) \end{aligned} \tag{13}$$

$$M_3 : \mathrm{DIST}(1)\mathrm{PURP}(2) + \mathrm{PURP}(1)\mathrm{ROUT}(0) + \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \quad (14)$$

$$M_4 : \mathrm{DIST}(0)\mathrm{PURP}(1) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{PURP}(2)\mathrm{USEF}(0) \leftrightarrow \mathrm{INST}(0) \quad (15)$$

$$M_5 : \mathrm{DIST}(0)\mathrm{PURP}(1) + \mathrm{DIST}(1)\mathrm{USEF}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) \leftrightarrow \mathrm{INST}(0) \quad (16)$$

$$M_6 : \mathrm{DIST}(0)\mathrm{PURP}(1) + \mathrm{DIST}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) \leftrightarrow \mathrm{INST}(0) \quad (17)$$

$$M_7 : \mathrm{PURP}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{ROUT}(0)\mathrm{USEF}(0) + \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \quad (18)$$

$$M_8 : \mathrm{PURP}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{PURP}(2)\mathrm{USEF}(0) + \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \quad (19)$$

$$M_9 : \mathrm{DIST}(1)\mathrm{USEF}(0) + \mathrm{PURP}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \quad (20)$$

$$M_{10} : \mathrm{DIST}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(1)\mathrm{ROUT}(0) + \mathrm{PURP}(2)\mathrm{ROUT}(1) + \mathrm{DIST}(0)\mathrm{ROUT}(1)\mathrm{USEF}(1) \leftrightarrow \mathrm{INST}(0) \quad (21)$$

While two PIs were essential in the analysis of the presence of the outcome mentioned earlier, as a result of which at least two causal paths were correctly identified although the overall model was incorrect given the information we had, no such PIs exist in the second analysis of the negation of the outcome. We have simply been lucky that the data-generating model $M^{\#}$ in Equation 10 (M_3 in Equation 14) and the incorrect model M_1 in Equation 11 (M_1 in Equation 12) had one path in common. If, for instance,

M_7 in Equation 18 had been assumed to be the correct model, there would have been no common paths at all.

The key device of QMC in the selection of the minimal sum models presented in Equations 9 and 11 is the *PI chart*. Table 5 depicts this chart for the solution containing the models in Equations 8 and 9 in the top part and for the solution containing the models in Equations 12 to 21 in the bottom part. It is customary to list the PIs along the rows and the configurations to be covered by the PIs along the columns. With respect to the presence of an institutional use of evaluations, there are five PIs and five configurations, while the chart shows nine PIs for the same number of configurations with regard to the absence of an institutional use. A cross, "$\times$", signifies that a PI includes a configuration, a dash, "$-$", shows this not to be the case.

As for INST(1)—the presence of an institutional use—PURP(0) and DIST(1)PURP(1) are the only PIs covering configurations 0001, 1011, and 1111, respectively, hence the need to include them. They are essential and the only columns they cover denote the so-called *distinguished configurations*. However, if PURP(0) belongs in any solution, DIST(0)USEF(0) becomes redundant because the former also covers configuration 0010. Left with configuration 0201 to be covered, the reader can verify that both DIST(0)PURP(2) and PURP(2)ROUT(0)USEF(1) make perfect alternatives, none in any way superior to the other. The former only contains one fewer "input to the switching circuit" than the latter but such considerations are undoubtedly ill-placed in the context of causal data analysis.

While the solution for the presence of an institutional use at least contains two essential PIs that feature in each model, the solution for its absence does not have this characteristic. Not a single PI is essential. How, then, does QMC arrive at M_1 in Equation 11? To understand this, it is necessary to introduce another central rule the algorithm uses: *row dominance*. In brief, one PI P_1 dominates another PI P_2 if all configurations covered by P_2 are also covered by P_1 but both PIs are not interchangeable (cf. McCluskey, 1965, p. 150). It can be shown that if row dominance is applied after all essential PIs have been chosen, the resulting model can never comprise more PIs than any other minimal model derived without row dominance. In Table 5, DIST(0)PURP(1) dominates PURP(1)ROUT(0) and DIST(0)ROUT(1)USEF(1), while DIST(1)PURP(2) dominates all the other five PIs, yielding M_1 in Equation 11 in consequence.[22]

How frequent are model ambiguities in empirical research? The results of a reanalysis of all applied articles listed in Table 1 for which data were available suggest that of 26 analyses that could have been performed for each possible outcome, 17 showed ambiguities in the range of 2 to 66

Table 5. Prime Implicant Chart for Outcomes INST(1) and INST(0).

		Configuration: DIST(·)PURP(·)ROUT(·)USEF(·)									
	Prime Implicant	0001	0010	0201	1011	1111	0101	0111	1200	1210	1211
INST(1)	PURP(0)	×	×	–	×	–	Essential and dominating				
	DIST(0)PURP(2)	–	–	×	–	–	Inessential				
	DIST(0)USEF(0)	–	×	–	–	–	Inessential				
	DIST(1)PURP(1)	–	–	–	–	×	Essential				
	PURP(2)ROUT(0)USEF(1)	–	–	×	–	–	Inessential				
INST(0)	DIST(0)PURP(1)	Inessential and dominating					×	×	–	–	–
	DIST(1)PURP(2)	Inessential and dominating					–	–	×	×	×
	DIST(1)ROUT(0)	Inessential					–	–	×	–	–
	DIST(0)USEF(0)	Inessential					–	–	×	×	–
	PURP(1)ROUT(0)	Inessential					×	–	–	–	–
	PURP(2)ROUT(1)	Inessential					–	–	–	×	×
	PURP(2)USEF(0)	Inessential					–	–	×	×	–
	ROUT(0)USEF(0)	Inessential					–	–	×	–	–
	DIST(0)ROUT(1)USEF(1)	Inessential					–	×	–	–	–

Note. distinguished configurations. × relations of row dominance.

models.[23] The only two computer programs that are currently capable of bringing this information to the attention of the researcher are the *QCA* package (Duşa & Thiem, 2014; Thiem & Duşa, 2013b, 2013c) and the *CNA* package (Ambuehl, Baumgartner, Kauffmann, & Thiem, 2014; Baumgartner, 2009, 2013), both for the *R* environment (*R* Development Core Team, 2014). To summarize, restrictions to minimal sum models are inappropriate for purposes of causal data analysis. Users should therefore employ software that is capable of identifying the full model space.

Conclusion

This article has set out to raise evaluation researchers' awareness of six potential pitfalls of QCA grouped under the following three aspect headings: case numbers, necessity relations, and model ambiguities. Although these make up but a fraction of possible stumbling blocks, they were chosen because of their pervasiveness in both methodological and applied work. First, it has been argued that numbers of cases are irrelevant to preferring QCA to other methods in general and to deciding on any of its variants in particular. Second, it has been shown that tests of simple conditions and preconceived disjunctions of simple conditions can provide only limited insights into the existence of minimal necessity relations. Furthermore, it has been demonstrated that the concept of necessity pervades QCA's analytical moment, contrary to what has been suggested in much of the methodological literature. Third, the article has sought to make out a strong case for scientific transparency by recognizing and reporting model ambiguities. In this connection, it has also identified serious problems with QMC optimization in deriving QCA solutions, most centrally the objective of deriving minimal sum models by means of row dominance.

Unfortunately, many of these pitfalls are not the consequence of slipshod applications of QCA in applied research, but they have in large part been homegrown by prematurely composed "standards of good practice" whose proposers have, in the same breath, paradoxically acknowledged the methodological immaturity of QCA. If applied users beware of the pitfalls introduced in this contribution, and provided that they avoid a mechanistic adherence to the procedural protocols enshrined in these standards given the current stage of methodological development, then research with QCA will gradually increase in quality, thereby creating a more solid foundation for cumulative knowledge generation and well-informed policy decisions.

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Notes

1. Albeit comprehensive, this list makes no claim to exhaustiveness. All information was collected from the COMPASSS website at http://www.compasss.org/bibdata.htm, July 3, 2014.
2. Other variants are temporal qualitative comparative analysis (QCA), a version of crisp set QCA (Caren & Panofsky, 2005; Ragin & Strand, 2008), and generalized-set QCA (gsQCA), which integrates multivalue QCA (mvQCA) and fuzzy set QCA (fsQCA; Thiem, 2014b).
3. More precisely, it is the number of configurations that matters in QCA, not the number of cases. Low case numbers may of course create problems of limited diversity, but this is an empirical problem, not a methodological one.
4. The same is true for almost all other areas of the social sciences in which QCA has made inroads, including political science and sociology. Only scholars in management and organization have rarely subscribed to the argument that the number of cases justifies the choice of QCA.
5. I do not include gsQCA (Thiem, 2014b) in this list since it may still be unknown to the vast majority of readers.
6. In contrast to Schneider and Wagemann (2010), Marx, Cambré, and Rihoux (2013, p. 29f.) argue that the analysis of necessity should follow that of sufficiency.
7. It remains unclear what the objective of analyzing such necessity relations without the concurrent analysis of sufficiency is. If it is causal data analysis, this separation is futile since this requires an integration of sufficiency and necessity relations. For the sake of the argument, however, I assume that scholars have some interest in minimal necessity relations independent of causal interpretation.
8. If some condition Ψ is necessary for some outcome Ω, then every disjunction $\Psi \vee \Phi$ is also necessary for Ω.
9. Since every statement of sufficiency with respect to some outcome can be converted into a statement of necessity for the negation of the same outcome, this cut-off must be equally applicable to isolated tests of necessity.
10. Cooper, Glaesser, and Thompson (2014) have also shown that it is extremely problematic to use the results of these isolated necessity tests to inform analyses of truth tables in order to derive QCA solutions.
11. By implication, equifinal QCA models that represent disjunctions of minimally sufficient conditions that are together necessary for an outcome should not be interpreted if no single higher-order concept that unites all minimally sufficient conditions comes to the mind of the researcher. Obviously, this reasoning is problematic, to say the least.

12. It is surprising that prominent QCA textbooks such as Rihoux and Ragin (2009) or Schneider and Wagemann (2012) introduce the implication operator "$\rightarrow$" but not the equivalence operator "$\leftrightarrow$."
13. McCluskey (1965, p. 123) has referred to these expressions as *complete sums*.
14. Such equivalences are more common in empirical applications than is often suggested. The results of a reanalysis of all applied articles listed in Table 1 for which data were available shows that of 26 analyses that could have been performed for each possible outcome, 20 involved the minimization of equivalences. This calculation includes Balthasar (2006; 10 outcomes), Gross and Garvin (2011; 6 outcomes), Holvoet and Dewachter (2013; 4 outcomes), Lam and Ostrom (2010; 4 outcomes), and Ozegowski (2013; 2 outcomes).
15. This section draws in large parts from joint work with Michael Baumgartner (Baumgartner & Thiem, 2014).
16. Many more such examples are provided in the replication file to this article.
17. The authors also report the parsimonious solutions for each of their two primary outcomes (note 10, p. 541), where they fail to list a second alternative model with respect to high performance on stimulating accountability, and the parsimonious solutions for the negation of the primary outcomes (notes 11 and 12, p. 542), where they fail to list 13 alternative models with respect to low performance on stimulating accountability and one model with respect to low performance on stimulating networking.
18. McCluskey (1965, p. 79) has referred to these expressions as *canonical sums*.
19. The variant of mvQCA has not found favor with many methodologists (Schneider & Wagemann, 2012, pp. 255–263), but unjustifiably so as Thiem (2013, 2014a, 2014b) shows. I use Balthasar's study because its data and results are ideally suited for demonstration purposes.
20. The other four outcomes are the conceptual use, the symbolic use, the process-related use and the general use of the evaluation by the evaluee. See Balthasar (2006, pp. 357–362) and Balthasar (2009) for more details.
21. These statistics are not listed again for reasons of space.
22. One reviewer remarked that only minimal sum models respect the principle of Occam's razor. Applying Occam's razor to QCA models, however, would imply that causal structures with fewer alternative paths, that is, lower equifinality, are significantly more common than causal structures with more paths. This is a very strong assumption which cannot simply be introduced by fiat but would first have to be corroborated.
23. This includes all studies and outcomes mentioned in note 14.

Supplemental Material

The online appendices are available at http://er.sagepub.com/supplemental.

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